

Efficient In-Context Learning for 3D Medical Image Segmentation

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Efficient In-Context Learning for 3D Medical Image Segmentation

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Declaration

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Furthermore, I declare that I have acknowledged the work of others by providing detailed references of said work.

I also hereby declare that my thesis has not been prepared for another examination or assignment, either in its entirety or excerpts thereof.

Freiburg, September 30, 2026

Place, date

Tidiane Camaret Ndir

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Abstract

Zusammenfassung

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1 Motivation

In-context learning (ICL) treats segmentation as a few-shot matching problem. The model receives a target scan and a small set of context pairs, each an image with a mask of the structure of interest, and predicts the corresponding mask for the target without task-specific fine-tuning. A single trained model can therefore segment new anatomical classes without retraining. This is useful for structures that are rare, newly defined, or annotated at only a few sites.

Most ICL segmentation models were developed and evaluated on 2D images. CT and MRI are volumetric, however, and processing them slice by slice discards through-plane context that helps to resolve thin or elongated structures. Working directly on 3D volumes avoids this, but makes compute a central constraint. The number of tokens grows with the cube of the volume's side length, and the cost of dense attention between target and context grows with the square of the number of tokens. Going from a 64^3 to a 256^3 volume multiplies the number of voxels by 64 and the cost of dense attention by roughly 4000.

This thesis addresses three questions that arise when building an efficient 3D ICL segmentation model.

1. **How should image and label information be combined?** In existing models, image and mask information is merged once, early in the network, either by concatenating channels before the encoder or through a single fusion operation. We instead keep image and label tokens as separate streams and let them exchange information at every layer through *dual attention* (subsection 3.2).
2. **Where should compute be spent?** Medverse [10] refines its prediction by processing the entire volume with a sliding window at every resolution step, including regions that the previous step has already ruled out. We use a *coarse-to-fine cascade* in which each level processes only the region localized by the previous level and receives that level's prediction as a prior (subsection 3.3).
3. **Where can training data come from beyond the annotated classes?** Real image-mask pairs cover only the classes and appearance range of the training cohort. We generate additional training tasks with a *synthetic task generator* that paints procedurally generated shapes with spatially correlated texture onto real scans, which gives exact labels for structures outside the annotated classes (subsection 3.4).

talk about query prior injection instead of localized processing

Contributions.

- A dual attention mechanism that keeps image and label tokens separate. We compare it with a compute-matched variant that merges both into a single token per cell, and with a dual U-Net cross-attention model initialized from Medverse.
- A coarse-to-fine approach that passes each level’s prediction to the next level as a prior and restricts fine-level processing to the localized region. We compare predicted priors with perturbed ground-truth priors, and compare accuracy and latency with Medverse’s coarse-to-fine approach.
- A synthetic task generator that paints procedural shapes onto real CT and MRI scans. We evaluate it against training on real data only and study its effect on real tasks.
- An evaluation on nine datasets covering unseen tasks such as lesions, a non-anatomical target region, and MRI as a held-out modality.

Outline. section 2 introduces in-context segmentation, the volumetric encoders and attention mechanisms the model builds on, and positions related methods with respect to the three questions above. section 3 describes the model. section 4 evaluates each component in turn. section 5 discusses the findings and their limitations.

2 Fundamentals

2.1 In-Context Segmentation

Given a target image I^t and a context set $\{(I_k^c, L_k^c)\}_{k=1}^K$ of image–mask pairs showing the same class, in-context segmentation predicts a target mask $\hat{L}^t$ in a single forward pass, without gradient updates or class-specific parameters. The setting is well-suited to medical imaging : First, medical annotation requires scarce domain expertise and is prohibitively expensive. Second, new anatomical structures and pathologies are defined much faster than large-scale supervised datasets can be curated for them. Finally, significant differences in acquisition protocols often limit how well a closed-set model trained at one site transfers to another.

2.2 Volumetric Encoders and Attention

Convolutional volumetric backbones. The 3D U-Net [4] architecture encodes a volume into a feature pyramid with a downsampling encoder and decodes it back to full resolution. Deeper stages have lower spatial resolution and more channels, while shallow stages keep spatial detail at a low channel count. Our model uses this split directly: the coarse, high-channel stages feed the attention layers, and the fine stages are passed to the decoder as skip connections (subsection 3.1). UNETR [8] is an earlier example of combining transformer attention with a convolutional decoder for 3D segmentation.

Attention. Scaled dot-product attention [19] computes, for each query, a weighted average of value vectors. The weights are given by the similarity between the query and each key:

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V, \quad (1)$$

where d is the key dimension. Multi-head attention runs several attention operations in parallel on lower-dimensional projections and concatenates their outputs, so that different heads can capture different relations within the same input.

Position encoding. Attention is permutation-invariant, so positional information has to be added explicitly. Rotary position embedding (RoPE) [17] rotates each query and key vector by an angle proportional to its position before the dot product. The resulting attention score depends only on the relative position of the two tokens. In 3D, RoPE can be applied per axis: the head dimension is split into three parts, and each part is rotated according to the token’s coordinate along one spatial axis. If these coordinates are given in physical units, i.e. scaled by the voxel spacing, attention between tokens

from volumes of different resolution reflects their physical distance rather than their grid indices. Our coarse-to-fine approach relies on this property, since its levels operate at different spacings (subsection 3.3).

2.3 Synthetic Training Data

Domain randomization trains a model on procedurally varied synthetic data so it does not overfit to the narrow appearance distribution of a single acquisition protocol or training cohort. SynthSeg [2] demonstrated this at scale for medical segmentation: training pairs are generated by sampling a random Gaussian-mixture appearance conditioned only on a label map, so the resulting segmentation network never sees a real image of any specific contrast during training yet generalizes across MRI contrasts and resolutions.

2.4 Related Work

Prior in-context segmentation models make different choices on the three questions raised in section 1. Table 1 summarizes them.

UniverSeg. UniverSeg [3] was one of the first in-context segmentation models for medical images. It operates on 2D images. Its CrossBlock combines target and context features with a CrossConv layer, a convolution applied to each concatenated target–context feature pair, at every scale of a U-Net-like network. It incorporates synthetic tasks (procedurally generated geometric shapes) alongside real medical datasets during training to improve zero-shot generalization.

Medverse. Medverse [10] uses two separate 3D U-Nets, one for the target and one for the context. The context branch receives image and mask concatenated as channels, while the target branch receives only the image. The two branches exchange features at several stages: the context queries the target in the encoder, and the target queries the context in the decoder. Following UniverSeg, it also relies on procedural synthetic shapes during training to expand task diversity.

Iris. Iris [7] encodes the context pairs once into a compact task embedding per class, using pixel shuffle operations in its context encoder. The target interacts with the context only through this embedding, via a single bidirectional cross-attention step in the mask decoder, and the embedding is reused across queries. Large volumes are processed via a sliding window approach, without refinement across scales.

Coarse-to-fine approaches. Deep Neural Patchworks [16] segments large volumes by stacking levels of increasing resolution. It is a supervised, closed-set method rather

say
what
it moti-
vates us
to gen-
erate
tasks
on real
scans

focus
more
on the
single
img-lbl
fusion
point

Table 1: Design choices of 3D in-context segmentation models on the three questions of section 1.

	Image-label fusion	Coarse-to-fine prior	Synthetic tasks
Iris	Before attention (pixel shuffle)	No	No
Medverse	Before attention (channel concatenation)	Ground truth mask	Procedural shapes
Ours	dual attention at every layer	Previous level prediction	Procedural shapes, painted on real scans

than an in-context one, but it is an early example of a coarse-to-fine cascade for 3D segmentation. Medverse brings a similar idea to in-context segmentation with next-scale autoregressive prediction, following next-scale image generation [18]. The first step predicts the whole volume at low resolution. Each later step doubles the resolution and processes the entire volume with a sliding window. The previous prediction is passed, together with the target image, through a third U-Net branch that shares weights with the context branch. During training, a downsampled ground-truth mask stands in for this prediction.

3 Methodology

3.1 Problem Setup and Backbone

We use the setting of subsection 2.1: given a target volume I^t and K context pairs (I_k^c, L_k^c) showing the same class, the model predicts the target mask $\hat{L}^t$. Prediction runs through a cascade of M levels with decreasing voxel spacing $r_1 > r_2 > \dots > r_M$. Each level predicts a mask, which is passed to the next level as a prior (subsection 3.3). Within a level, the model consists of a shared encoder, a stack of dual attention layers, and a decoder (Figure 1).

At every level, all volumes are resampled onto the same grid of 128^3 voxels, and only the voxel spacing changes between levels. A coarse level therefore covers a large field of view at low resolution, and a fine level a small field of view at high resolution: at 6 mm spacing the grid spans 768 mm per side, at 1.5 mm only 192 mm. During training, the spacing is sampled from a range rather than fixed to the levels used at inference (subsection 3.3).

Give the training spacing range in the results settings

Shared encoder. All $K + 1$ images of a task pass through the same encoder, a plain convolutional encoder in the style of nnU-Net [11] with four stages, trained from scratch with He initialization. The encoder sees only images; masks enter the model through the label tokens described below. The first two stages (0 and 1) are the fine stages: they keep their own resolution and serve as skip connections for the decoder (subsection 3.2). The last two stages (2 and 3) are the coarse stages. Their features are resampled to a common grid of $R \times R \times R$ cells with $R = 16$, giving $N = R^3 = 4096$ cells per volume, and concatenated along the channel dimension. Each cell therefore covers $p^3 = 8^3$ voxels of the input grid, with $p = 128/R$. The attention layers always operate on this fixed grid, while the decoder can still produce a mask at the full 128^3 resolution.

subsection 4.2 says the fusion arms warm-start the encoder and decoder from a shared checkpoint. Is that a checkpoint of this model, itself trained from scratch? If so, say so there.

Draft diagram, ported from the conference draft; finalize labels/scale once the architecture is settled.

Tokens. Each cell i of each volume v is represented by two tokens of dimension $D = 768$. The image token is a linear projection of the concatenated coarse features at

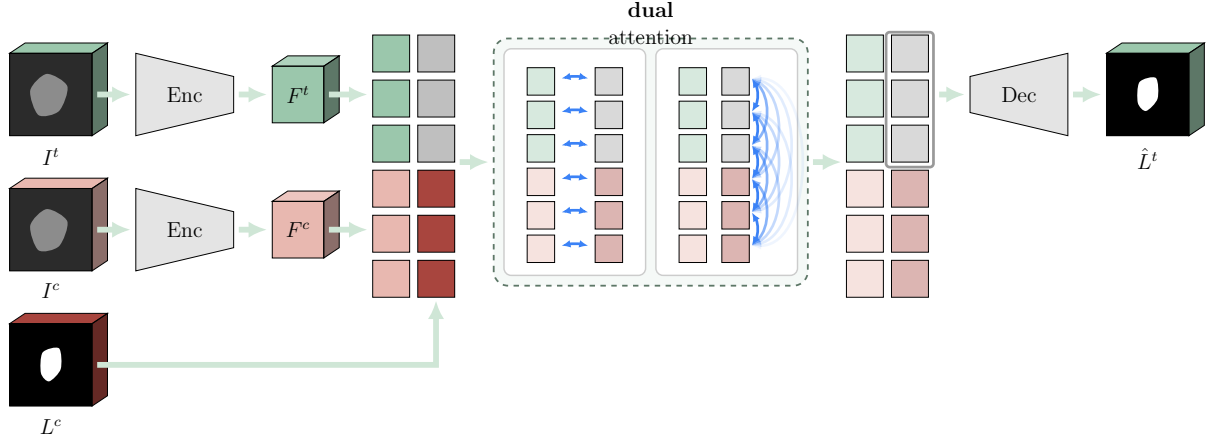


Figure 1: Overview of the model at one cascade level. Target and context images pass through a shared encoder; image and label tokens are processed by dual attention; a decoder produces the target mask.

that cell,

$$z_{v,i}^{\text{img}} = W_{\text{img}} f_{v,i}, \quad (2)$$

where $f_{v,i}$ denotes the resampled stage-2 and stage-3 features of cell i . The label token embeds the part of the mask that falls into the cell. We flatten the $p^3 = 512$ mask voxels of cell i into a vector $m_{v,i}$ and project it linearly:

$$z_{v,i}^{\text{lbl}} = \text{LblEmbed}(m_{v,i}) = W_{\text{lbl}} m_{v,i} + b_{\text{lbl}}. \quad (3)$$

For a context volume, $m_{v,i}$ is taken from the ground-truth mask L_k^c . For the target, the mask is unknown. At the first cascade level, every target cell receives the same placeholder: the average of $m_{v,i}$ over all cells of all context volumes, embedded with LblEmbed. It carries the average amount of foreground in the context masks, but no information about where the target is. At later levels, the target’s label tokens are computed from the previous level’s prediction instead (subsection 3.3).

Learned embeddings mark each token as belonging to the target or to a context volume. Eight learned register tokens, shared across all volumes, are prepended to the sequence. They are not tied to any cell and give the attention layers a fixed amount of global memory.

3.2 Dual Attention

The attention layers have two roles : combining image and label information, and matching the target against the context. In Medverse and Iris, these happen at fixed points in the network (subsection 2.4). Figure 2 contrasts Medverse’s design with the shared-encoder design used in this thesis.

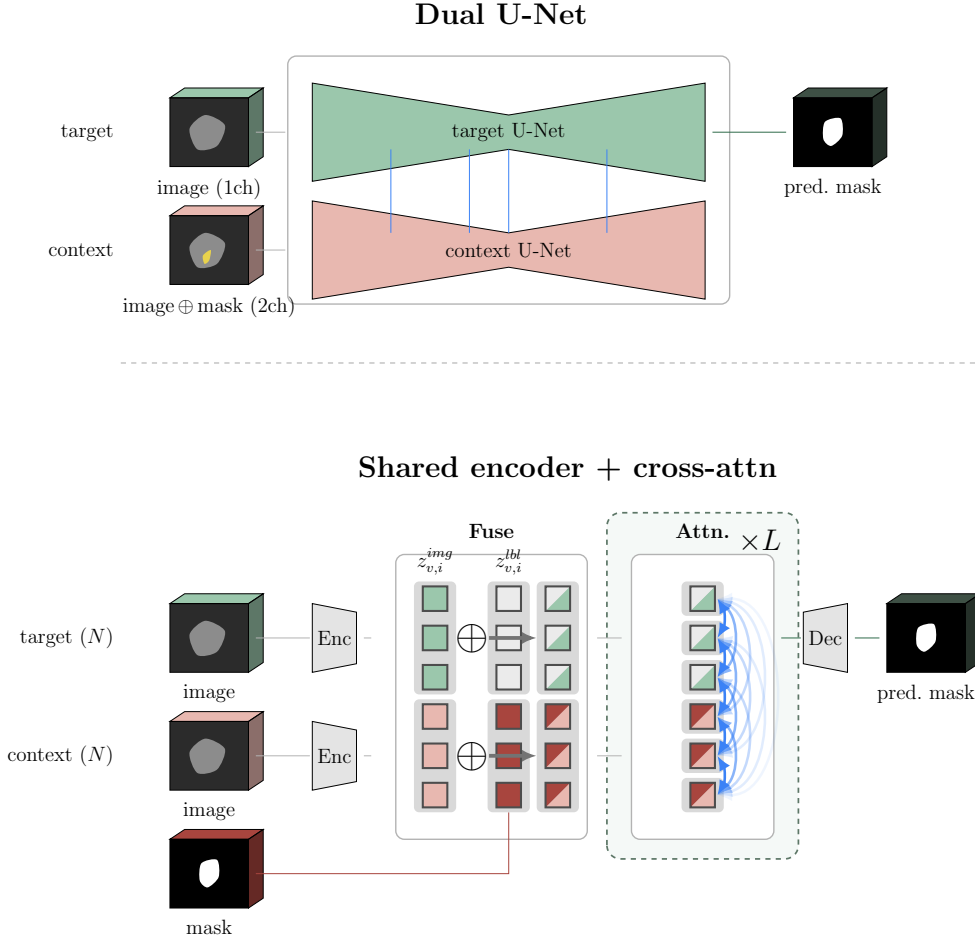


Figure 2: Two ways of coupling target and context, shown for one context volume ($K=1$) and three of its N tokens. Top: Medverse feeds the target image, and the context image and mask concatenated as channels, into two separate U-Nets that exchange features through cross-attention at several stages (blue arrows). Bottom: the shared-encoder design used in this thesis. Only images pass through the encoder. Label information is added to the encoded tokens afterwards, the tokens attend across cells and volumes for L layers, and a shared decoder produces the mask. Figure 3 shows the two ways of adding label information that we compare.

Token array. We arrange all tokens of a task in an array $Z \in \mathbb{R}^{S \times 2 \times D}$. Each of its S rows corresponds to one cell of one volume, and its two columns hold the image stream and the label stream (subsection 3.1). With $K+1$ volumes of N cells, $S = (K+1)N$. Dual attention attends along each of the two axes of this array in turn, following the general pattern of the dual attention network of Fu et al. [6], which combines position-wise and channel-wise attention for scene segmentation.

Confirm: register and pooling rows also have an entry in both streams (implied by the $c=2$ layout). How are they initialized in the label stream?

Layer structure. Each of the $L = 4$ layers applies three residual blocks, each preceded by RMS normalization:

$$Z \leftarrow Z + \text{Attn}_{\text{mod}}(\text{RMSNorm}_1(Z)), \quad (4)$$

$$Z \leftarrow Z + \text{Attn}_{\text{pos}}(\text{RMSNorm}_2(Z)), \quad (5)$$

$$Z \leftarrow Z + \text{MLP}(\text{RMSNorm}_3(Z)). \quad (6)$$

- **Modality axis** (Attn_{mod}). Within each row, the image token and the label token attend to each other. This is attention over a sequence of length two, computed independently for every row, so its cost grows linearly with S .
- **Position axis** (Attn_{pos}). Within each stream, the tokens of all rows attend to each other: image tokens attend to the image tokens of all cells, volumes, and registers, and label tokens likewise to label tokens. The two streams use the same attention weights but are processed as separate sequences of length S . Attention is unmasked, so every row attends to every other row, including context rows to target rows and vice versa. Queries and keys use 3D axial RoPE with physical coordinates (subsection 2.2), so attention depends on the physical distance between cells. The cost grows quadratically with S .
- **MLP**. Two linear layers with a GELU activation in between, applied to each token independently, with hidden dimension 3072.

All attention operations use 12 heads with a token dimension of $D = 768$.

Confirm: (1) the position axis shares its weights between the two streams (implied by batching the streams into the batch dimension); (2) how RoPE is applied to register and pooling rows, which have no spatial position.

The two streams exchange information only in the modality axis, which is local to a cell. Matching across cells and volumes happens within each stream. Because the modality axis comes first in every layer, the tokens entering the position axis already combine image and label information of their cell, so both streams can use both kinds of information when matching across cells. Since attention is unmasked, the context tokens also depend on the target: the context is encoded anew for each target, rather than once per task as in Iris.

Single-axis attention baseline. As a baseline, the single-axis attention variant (Figure 3, top) merges the image and label token of each cell into a single token before the attention layers, using a pixel shuffle–unshuffle operation similar to the one Iris uses in its context encoder. The array then has a single column, and each layer applies only the position axis and the MLP. The image and label information of a cell is thus combined once, before any attention, rather than in every layer. Without the modality axis and the second stream, a layer costs roughly half as much, so this variant uses $L = 9$ layers to match the compute of dual attention (subsection 4.2).

Describe the merge precisely: is the 8^3 mask patch unshuffled into channels and concatenated with the coarse features before the projection W_{img} ? Otherwise the same block structure (RMSNorm, MLP, 12 heads)?

3.3 Coarse-to-Fine Cascade

Each of the M levels processes a single 128^3 crop; levels differ only in spacing and hence field of view. The prediction of one level determines where the next crop is placed and initializes the target’s label tokens.

Crop placement. The first crop is centered at the center of the volume, with a spacing chosen per dataset to cover the region where the target can occur. Each later crop is centered at the probability-weighted centroid of the previous prediction, or at the volume center if that prediction is empty. The finest spacing is chosen so that the field of view exceeds the largest expected target (Table 7). Each level thus costs one forward pass, independent of volume and target size. We refer to this as region restriction. Its drawbacks are that a target missed at a coarse level cannot be recovered, and that a target with several distant components may not fit into one fine crop.

Stitching. The coarsest prediction is pasted into the original volume first; each finer level then overwrites the region covered by its crop.

Prediction as prior. At level $\ell > 1$, the target’s label tokens are computed from the previous prediction, cell by cell as in Equation 3:

$$z_{\ell,0}^{\text{lbl}} = \text{LblEmbed}(\text{Warp}_{\ell-1 \rightarrow \ell}(\hat{L}_{\ell-1}^t)), \quad (7)$$

where $\hat{L}_{\ell-1}^t$ is the sigmoid probability map of level $\ell-1$, detached from the computation graph, and Warp resamples it onto the crop grid of level ℓ . The prior thus enters the

Draft diagram, ported from the conference draft; finalize once the layer factorization is settled. The position-axis panel should show two separate attentions, one per stream, not one joint attention over im-

Table 2: Shape families. Target diameters are drawn from 10–50 mm unless stated otherwise.

Family	Family-specific parameters
Blob, disk	–
Splatter	2–6 components, spread 5–20 mm
Scatter field	10–120 components, spread 20–140 mm
Cylinder	length 30–100 mm
Vessel	trunk 40–120 mm, branching depth 2–4
Torus	ratio of major to minor radius 2–5

same stream as the context masks, instead of a separate branch as in Medverse (subsection 2.4); see Figure 4.

Training. All M levels are processed in one forward pass. The first crop is centered on the ground truth (at inference: on the volume center), later crops on the previous prediction, at its centroid or at a random foreground point. The coarsest spacing is drawn per batch, and each finer spacing log-uniformly between the previous spacing and a lower bound, with a minimum ratio between levels. The prior is either the model’s own prediction (predicted prior) or the downsampled ground-truth mask; subsection 4.3 compares both.

Values: M , spacing range, lower bound, minimum ratio, centroid vs. random-point probability, foreground threshold, perturbation magnitudes. Confirm inference always uses the centroid.

Draft diagram, ported from the conference draft; finalize once the number of levels M is settled.

3.4 Synthetic Task Generation

During training, a fraction p_{synth} of tasks is generated synthetically. For each synthetic task, one shape family and one set of shape and appearance parameters are drawn; these play the role of the class. Each of the $K + 1$ volumes then receives its own instance of the shape, painted onto its own image. Shape, size, and position vary between volumes with ratios of 0.3, 0.3, and 0.5, where 0 reuses the task-level value and 1 draws it independently.

Shapes. Masks come from seven procedural families (Table 2). The scatter field places many small components across a region to imitate multi-focal lesions, which few structures in the training cohort resemble. Sizes are defined in millimeters and converted to voxels at each cascade level, so a shape is the same physical object at every level.

Canvas. We compare two sources for the images. The *real-host* canvas uses real TotalSegmentator scans. A host structure is drawn, and $K + 1$ different subjects containing it are sampled; the shape is placed within ± 30 mm per axis of the host structure’s centroid. Since the volumes of a task are different scans, the target cannot be found by comparing it with the context images voxel by voxel. The *synthetic-bank* canvas uses a bank of label maps from segmented CT scans of about 21 datasets. A class is drawn, and $K + 1$ label maps containing it are selected, optionally preferring maps of similar geometry. As in SynthSeg [2], the image is synthesized from the label map with Gaussian-mixture intensities, and the shape is painted into it, with an optional blend of core and rim intensity. Only the anatomical layout of this canvas is real; its appearance is synthetic.

Appearance on the real-host canvas. The shape is painted relative to the local tissue statistics μ and σ of the host image:

$$I(x) = \mu + c \sigma + \sigma n(x), \quad c \sim \mathcal{U}(-2.5, 2.5), \quad (8)$$

with the contrast c drawn once per task. The range covers the lesion-to-tissue contrast we measured on seven of the evaluation datasets (10th to 90th percentile: -0.87 to 2.00), widened symmetrically because the sign of the contrast is not known for an arbitrary host. The texture $n(x)$ is fractal value noise with unit variance: four octaves of random grids, starting from 15 mm cells and halving the cell size per octave, weighted by 0.5^k .

Over which region are μ and σ computed (a ring around the shape?) Which seven datasets were used for the contrast calibration?

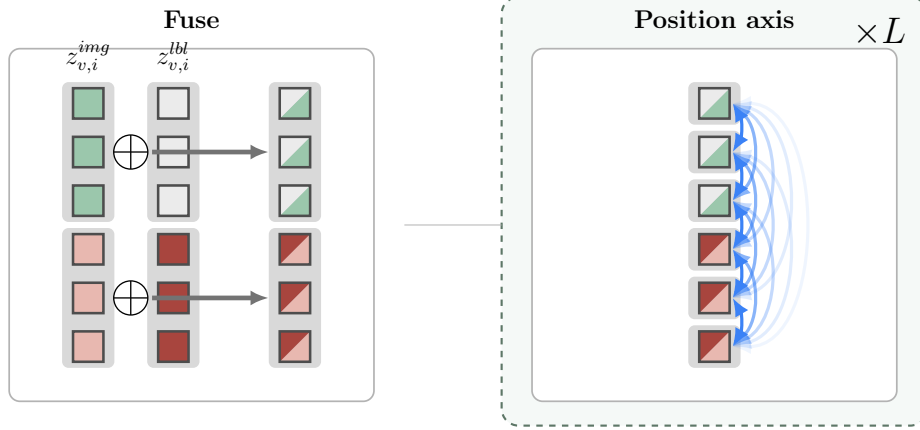
3.5 Training Objective

Each cascade level is supervised at its own resolution with the sum of a binary cross-entropy and a soft Dice loss:

$$\mathcal{L} = \sum_{\ell=1}^M \lambda_{\ell} (\mathcal{L}_{\text{BCE}}^{\ell} + \mathcal{L}_{\text{Dice}}^{\ell}), \quad (9)$$

with uniform weights $\lambda_{\ell} = 1$. Because the prior is detached (subsection 3.3), the loss at level ℓ does not propagate into level $\ell-1$. The single-level experiments of subsection 4.2 use $M = 1$.

Single axis attention



Dual attention

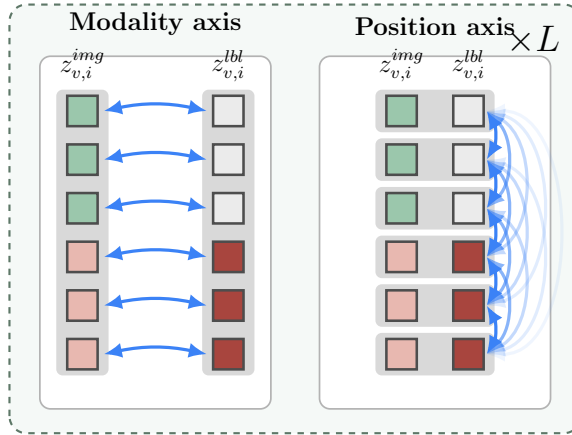


Figure 3: The two variants of the shared-encoder design compared in subsection 4.2, shown for one context volume and three of its N cells. Top: early fusion (baseline). Image and label tokens are merged once ($\oplus$), and the fused tokens attend across cells and volumes for L layers. Bottom: dual attention. Image and label tokens stay in separate streams. In each of the L layers, the two tokens of a cell attend to each other (modality axis), and then each stream attends across all cells and volumes (position axis). The target’s label token (gray) holds the prior of subsection 3.3, not ground truth.

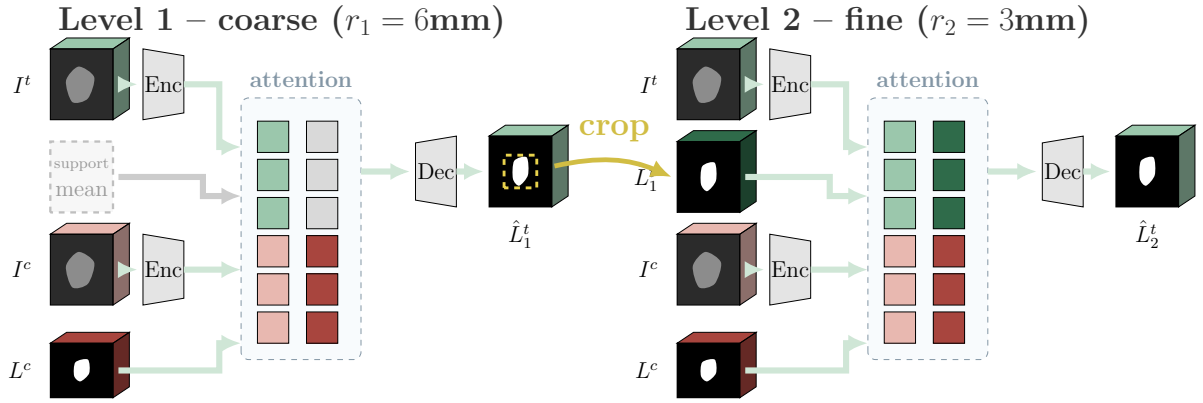


Figure 4: Label prior across two cascade levels. At level 1, the target’s label tokens hold a placeholder (dashed); at level 2, a crop of the level-1 prediction $\hat{L}_1^t$.

4 Results

4.1 Experimental Setup

Training data. Models are trained on the CT part of TotalSegmentator [20] (1082 training, 57 validation, and 89 test subjects) and, except in subsection 4.2, on the TotalSegmentator MRI release [5] (561 training subjects from University Hospital Basel and the NIH Imaging Data Commons). Training uses 61 of the 117 classes, chosen to balance anatomical categories and structure thickness: 12 abdominal and pelvic organs, 8 thoracic, head, and spinal organs, 5 muscles (one side of each pair), 6 limb, shoulder, and pelvic bones, 11 vertebrae spanning all spine regions, 9 ribs and sternal structures, and 10 vessels. The remaining 56 classes form the unseen split. (Table 14) It includes structures close to training classes, such as the contralateral side of each training muscle. On MRI, 31 of the 61 training classes are available.

Model configurations. The experiments were run at different stages of development, so the three results sections use different model configurations (Table 3). Numbers from different sections are therefore not directly comparable. In particular, TotalSegmentator MRI is a held-out modality for the fusion experiments, which train on CT only, but part of the training data in the other two sections.

Resolve before submission: report a single final configuration, or keep this split?

Baselines. The external baseline is Medverse [10] with its released weights (subsection 4.3, subsection 4.4). In subsection 4.2, we additionally train Medverse’s architecture from its released weights with our training recipe.

No Iris baseline exists yet; the comparison in subsection 2.4 remains architectural, not empirical.

Evaluation protocol. Each target is segmented with a single context pair ($K = 1$). The context subject is drawn at random, with a fixed seed per target, from the same dataset and split as the target, among the subjects that contain the target class. Every subject that contains a class serves once as a target for that class. For each model, we use the checkpoint with the best Dice on the in-distribution validation split; no evaluation dataset is used for checkpoint selection.

See open points: MRI validation split, n in Table 4, K during training.

Table 3: Model configurations used in the results sections.

Experiment	Section	Training data	Levels	p_{synth}
Image–Label Fusion	4.2	TotalSeg CT	1	0
Cascade	4.3, 4.4	TotalSeg CT + MRI	2–3	0
Synth	4.4	TotalSeg CT + MRI	1	0.5
Synth + Cascade	4.4	TotalSeg CT + MRI	2–3	0.5

Metrics. Accuracy is reported as Dice, computed per class and averaged over classes unless stated otherwise, and as Normalized Surface Dice (NSD) with a tolerance of 3 mm. Compute is reported as GFLOPs per forward pass and as latency per sample. All training and evaluation runs on a single NVIDIA RTX PRO 6000 Blackwell Server Edition GPU (96 GB).

Precision of the latency measurements in Table 8 (fp32 as in Table 6?).

Datasets. Table 4 lists the evaluation datasets, grouped by their relation to the training classes. TotalSegmentator and FLARE22 contain classes from the TotalSegmentator vocabulary, so their classes split into seen and unseen ones. The other datasets contain targets that do not occur in training: new anatomical structures, lesions, a measurement region that is not an anatomical object, and, for the fusion experiments, an unseen modality. Because of this split, "held-out" is used at two levels below: at the dataset level (a dataset whose images were never trained on, e.g. FLARE22, whose classes are nonetheless mostly in-vocabulary) and, where stated explicitly as such, at the class level (excluding classes that are in the training vocabulary, even if their dataset is held out).

4.2 Image–Label Fusion

This section compares three architectures. The comparison between Dual U-Net Cross-Attn and early fusion tests the change from two U-Nets coupled by cross-attention to a shared encoder with attention layers. The comparison between early fusion and dual attention tests whether, within the shared-encoder design, keeping image and label tokens separate improves over merging them once.

Architectures. *Dual U-Net Cross-Attn* is Medverse’s architecture (subsection 2.4), initialized from the released weights and then trained with our recipe. It is therefore a different model from the released Medverse baseline used in subsection 4.3. *Early fusion* and *dual attention* are the two variants shown in Figure 3. Early fusion uses $L=9$ layers to match the compute of dual attention (subsection 3.2). This gives both variants

Table 4: Evaluation datasets, grouped by relation to the training classes. n : number of subjects; n_{cls} : number of evaluated classes.

Dataset	Modality	Target	n	n_{cls} (seen/unseen)
<i>Training vocabulary (seen and unseen classes)</i>				
TotalSegmentator [20]	CT	full-body anatomical structures	89	117 (61/56)
FLARE22 [13]	CT	abdominal organs	50	13 (11/2)
<i>New targets</i>				
ISLES22 [9]	MRI DWI	stroke lesion	250	1
Shifts-MS [14]	MRI FLAIR	MS lesions	46	1
MSD Hippocampus [1]	MRI T1	hippocampus	260	2
MSD Prostate [1]	MRI T2 + ADC	prostate zones	64	2
ATLAS v2.0* [12]	MRI T1	chronic stroke lesion	654	1
GNC_705	MRI Dixon	kidney lesions	610	13
NasalSeg [21]	CT	nasal cavity and sinuses	107	5
<i>Non-anatomical target</i>				
HU_LWK1	CT	center of L1 vertebrae	36	1
<i>Additional modality (unseen in subsection 4.2)</i>				
TotalSeg MRI [5]	MRI	full-body anatomical structures	47	50 (31/19)

*Obtained from an unofficial mirror; provenance not yet confirmed.

Table 5: Training settings shared by the three architectures of subsection 4.2.

Optimizer	AdamW
Learning rate	10^{-4}
Schedule	cosine decay, 10-epoch warmup
Weight decay	10^{-2}
Batch size	8
Gradient clipping	1.0
Loss	BCE + Dice
Epochs	250

similar compute (3973 vs. 3892 GFLOPs). Parameter count could not be matched at the same time, and early fusion has 52% more parameters. Dual U-Net Cross-Attn is not compute-matched to either variant.

Training. All three architectures use the same training recipe (Table 5), data pipeline, and augmentations. They are trained on the CT cases and training classes of TotalSegmentator, without synthetic tasks, at a single level. Their initialization differs. Dual U-Net Cross-Attn starts from Medverse’s weights. Early fusion and dual attention are randomly initialized. Dual U-Net Cross-Attn completed its training budget; early fusion and dual attention were stopped before completing theirs, so the comparison also reflects this difference in training progress, not only initialization.

Table 6: Dice of the three architectures at a single level (3 mm), with parameter count, compute, and latency.

	Dual U-Net Cross-Attn	Early fusion	Dual attention
TotalSeg CT, seen	0.348	0.574	0.582
TotalSeg CT, unseen	0.221	0.408	0.426
FLARE22, seen	0.532	0.693	0.709
FLARE22, unseen	0.443	0.632	0.604
MSD Prostate	0.377	0.205	0.247
MSD Hippocampus	0.172	0.122	0.184
HU_LWK1	0.146	0.079	0.102
TotalSeg MRI	0.178	0.232	0.259
ISLES22	0.046	0.037	0.043
Shifts-MS	0.087	0.038	0.109
ATLAS v2.0	0.031	0.014	0.022
GNC_705	0.044	0.014	0.025
NasalSeg	0.434	0.272	0.358
Mean of 10 held-out sources [†]	0.187	0.174	0.201
Parameters	71.1M	74.1M	48.7M
GFLOPs	2362.6	3973.1	3892.5
Latency per sample	73.2 ms	73.0 ms	72.5 ms

[†]MSD Prostate, MSD Hippocampus, HU_LWK1, TotalSeg MRI, FLARE22 (all 13 classes, not split by seen/unseen as above), ISLES22, Shifts-MS, ATLAS v2.0, GNC_705, and NasalSeg — held-out at the dataset level (subsection 4.1). Sample-weighted: each dataset’s macro Dice is weighted by its number of evaluated tasks (36 for HU_LWK1 to 1490 for GNC_705) rather than counted once per dataset, since task counts differ by up to 40×. The unweighted mean-of-datasets (0.203/0.170/0.204) and the fully pooled per-task mean (0.186/0.172/0.200) rank the three architectures identically.

Evaluation. All three architectures are evaluated at a single level with 3 mm spacing, in fp32, using the best checkpoint of each. Table 6 reports the results, grouped as in Table 4.

Specify where the 3 mm crops are placed (centered on the ground truth?).

Results. Early fusion outperforms Dual U-Net Cross-Attn on the classes from the training vocabulary, by 0.16 to 0.23 Dice on TotalSegmentator and FLARE22, and by a smaller margin on the unseen-modality TotalSeg MRI (0.232 vs. 0.178). It is worse on the three sources with new or non-anatomical targets: MSD Prostate, MSD Hippocampus, and HU_LWK1.

Dual attention improves over early fusion on seven of the eight rows. The gains are small on the other in-vocabulary rows (0.008 to 0.018) and larger on the new targets and the unseen modality: MSD Hippocampus (+0.062), MSD Prostate (+0.042), TotalSeg

MRI (+0.027), and HU_LWK1 (+0.023). The exception is the unseen FLARE22 classes, where dual attention scores 0.028 lower.

Compared with Dual U-Net Cross-Attn, dual attention is better on six of the eight rows, including MSD Hippocampus (0.184 vs. 0.172), but remains worse on MSD Prostate (0.247 vs. 0.377) and HU_LWK1 (0.102 vs. 0.146).

On the five lesion and cavity datasets in Table 6 (ISLES22, Shifts-MS, ATLAS v2.0, GNC_705, NasalSeg), the pattern reverses: Dual U-Net Cross-Attn is best of the three on four of five, losing only on Shifts-MS, where dual attention is best (0.109 vs. 0.087). Dual attention’s margin behind Dual U-Net Cross-Attn on the other four is at most 0.076 (NasalSeg).

On the sample-weighted mean over all ten held-out sources, early fusion is below Dual U-Net Cross-Attn (0.174 vs. 0.187), while dual attention is above it (0.201); the unweighted mean-of-datasets and the fully pooled per-task mean rank the three architectures the same way. This aggregate advantage for dual attention comes from the first five sources, not the lesion and cavity ones, where Dual U-Net Cross-Attn wins more often. The shared-encoder design alone thus improves accuracy on the training vocabulary, and dual attention improves on held-out sources in aggregate, though not on every individual source.

Dual attention has the fewest parameters of the three (48.7M). Its compute is similar to early fusion by construction and 65% higher than Dual U-Net Cross-Attn. Latency is nearly identical for all three (72.5 to 73.2 ms) despite these differences in GFLOPs.

4.3 Coarse-to-Fine Cascade

This section evaluates the Cascade configuration (Table 3). We first examine how accuracy develops across levels and how well the coarse level localizes the target. We then compare with Medverse in accuracy and latency, compare the two sources of the label prior, and finally test a register-carry variant.

Accuracy across levels. When the crop is centered on the ground truth, reducing the spacing from 4 mm to 1.5 mm raises macro Dice on TotalSegmentator from 0.361 to 0.538 (117 classes, 3897 samples). Because these crops use the ground-truth position, this shows the benefit of finer resolution, not of the cascade. Table 7 therefore reports the Dice of each level when its crop is placed using the previous level’s prediction. Dice still increases from level to level on every dataset, for example from 0.603 to 0.718 to

Table 7: Dice of the Cascade model at each level, with crops placed using the previous level’s prediction, and Dice of the final stitched prediction.

Dataset	Spacings (mm)	Dice per level	Stitched
FLARE22	[6, 3, 2.5]	0.603 → 0.718 → 0.751	0.751
MSD Prostate	[3, 1.5, 0.75]	0.276 → 0.361 → 0.381	0.375
HU_LWK1	[6, 3, 1]	0.065 → 0.109 → 0.152	0.152
TotalSeg MRI	[6, 3, 1.5]	0.365 → 0.479 → 0.495	0.503
NasalSeg	[1.5, 0.8]	0.484 → 0.547	0.547

0.751 on FLARE22.

This shows that the cascade works without ground-truth localization, but not that it beats a single fine-resolution pass over the whole volume. The planned –region-restriction ablation answers this.

Outside the finest crop, the stitched mask contains the predictions of coarser levels, so its Dice can be higher than the finest level’s (TotalSeg MRI), lower (MSD Prostate), or unchanged (HU_LWK1, FLARE22, NasalSeg) when the target lies entirely inside the finest crop and there is nothing outside it for a coarser prediction to add.

Localization. On TotalSeg CT (117 classes), at the coarsest level, the crop of the middle level, placed at the centroid of the coarse prediction, contains on average 92.2% of the target’s voxels; placed at the ground-truth centroid, the same crop contains 97.0%. Across classes, this containment correlates only weakly with the Dice gain from the finer levels ($r = -0.22$, $p = 0.018$, $n = 117$, $r^2 \approx 0.05$). This measures only the coarsest-to-middle transition, not the middle-to-finest one. Since most of the target already lies inside the box at this first step, most of the remaining error at this step arises within the fine crops rather than from localization; whether the same holds for the middle-to-finest transition is not tested here.

Comparison with Medverse. Table 8 compares Ours with Medverse’s released weights, using its own autoregressive coarse-to-fine inference with the number of steps matched to our number of levels per dataset (Medverse’s native, unmatched depth is compared separately in ??). Four lesion datasets (ISLES22, Shifts-MS, ATLAS v2.0, GNC_705), where both numbers are below 0.1, appear at the bottom of the table. TotalSeg CT is not included in this comparison (see the table’s \todo). MSD Hippocampus is processed at a single level, so its row compares the base models without a cascade.

Table 8: Ours vs. Medverse: macro Dice and mean latency per sample. Medverse is evaluated with its number of autoregressive steps matched to our number of levels per dataset.

Dataset	Spacings (mm)	Dice		Latency (ms)	
		Ours	Medverse	Ours	Medverse
FLARE22	[6, 3, 2.5]	0.751	0.359	311	4894
MSD Prostate	[3, 1.5, 0.75]	0.375	0.357	279	5006
MSD Hippocampus	[0.5]	0.303	0.686	99	50
HU_LWK1	[6, 3, 1]	0.152	0.020	439	5336
TotalSeg MRI	[6, 3, 1.5]	0.503	0.055	194	5188
NasalSeg	[1.5, 0.8]	0.547	0.737	161	1886
Mean (sample-weighted) [†]		0.526	0.393	206	3491
ISLES22	[3, 1.5]	0.069	0.059	155	1894
Shifts-MS	[3, 1.5]	0.031	0.098	289	1911
ATLAS v2.0	[3.8, 1.9]	0.026	0.024	149	1862
GNC_705	[6, 3, 1.2]	0.024	0.007	288	4874

[†]Dice and latency are both sample-weighted, each dataset weighted by its own number of evaluated tasks (36 for HU_LWK1 to 650 for FLARE22). Ours and Medverse are both averaged over the same six datasets above the second midrule (Ours: Dice 0.526, unweighted 0.439; latency 206 ms, unweighted 247 ms. Medverse: Dice 0.393, unweighted 0.369; latency 3491 ms, unweighted 3727 ms). The four lesion datasets below the second midrule are excluded from both means (both models score below 0.1 on every one of them).

TotalSeg CT is commented out of this table (and of Table 7 and ??) — decided out of scope for the final paper. Its numbers, where they exist, are still available if this changes: see `TOD0.md` for what is and isn't archived (our own model's TotalSeg CT cascade run is done; Medverse's matched and native cells there never completed).

Against Medverse with matched steps, Ours is more accurate on seven of the ten datasets: FLARE22, MSD Prostate, HU_LWK1, TotalSeg MRI, ISLES22, ATLAS v2.0, and GNC_705. Medverse is more accurate on MSD Hippocampus, reportedly part of its training classes but not of ours, on NasalSeg, which neither model saw during training, and on Shifts-MS.

Figure 5 shows one case from each side of this comparison, using the context, ground truth, and prediction of both models on the same target subject (Medverse run in its native-autoregressive mode, not the step-matched mode of Table 8; see ??). Ground truth is shown in green, Ours in blue, Medverse's in orange throughout this section. On FLARE22 gallbladder, Ours recovers the organ's shape (Dice 0.79) while Medverse predicts a small mislocated fragment (Dice 0.00). On MSD Hippocampus, the pattern reverses: Ours is fragmented into disconnected blobs (Dice 0.34), while Medverse

FLARE22 gallbladder (subject FLARE22_Tr_0001)

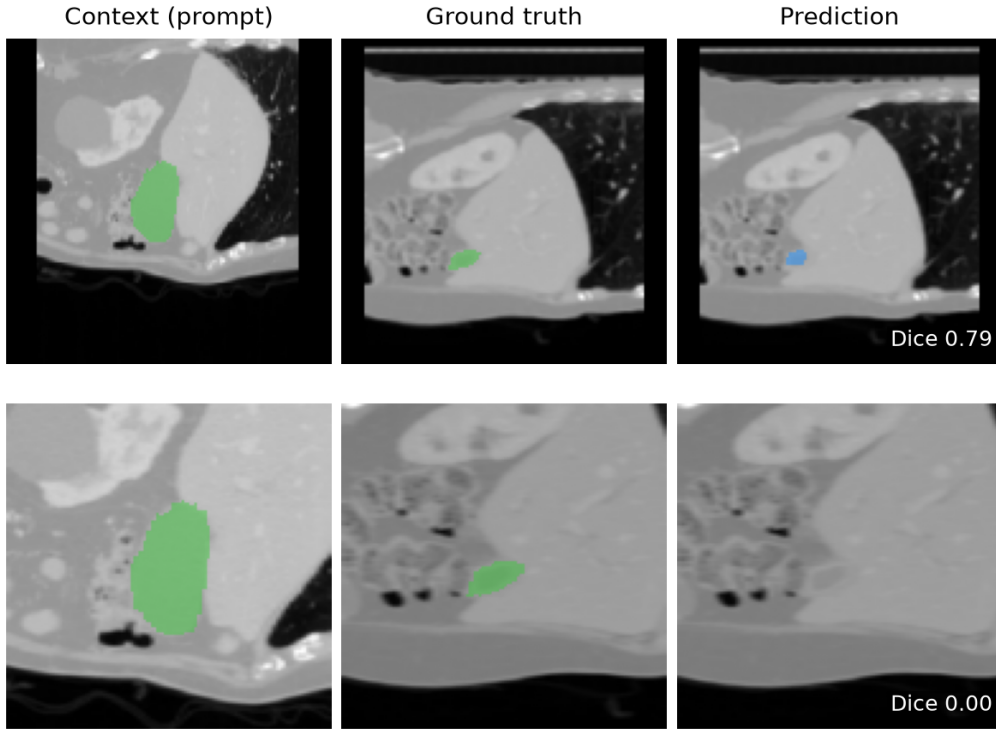


Figure 5: FLARE22 gallbladder, subject FLARE22_Tr_0001: Ours recovers the organ shape where Medverse’s native-autoregressive prediction misses almost entirely.

produces a single contiguous region closer to the ground-truth shape (Dice 0.70).

Both panels are re-rendered directly from `run_cascade / model.predict()` outputs on these two (target, context) pairs – checkpoints and settings matching 2026-09-26_147_cascade_randomfg_predprior_regon_flare22_ood / 2026-09-15_medverse_ar_flare22 and 2026-09-24_146_cascade_randomfg_predprior_regoff_msd_hippocampus_ood / 2026-09-14_medverse_ar_msd_hippocampus – rather than the harness’s own saved diagnostic panels, so the color scheme could be made consistent across models (green/blue/orange). Each sample is the first the eval loader hit for its class, not a systematic best/worst search.

Ours is about $17\times$ faster than step-matched Medverse, sample-weighted over the six datasets in Table 8 (206 vs. 3491 ms per sample; unweighted, $15\times$: 247 vs. 3727 ms). Our model runs one forward pass per level, whereas Medverse slides a window over the whole volume at each step; this difference and the architectures are not separated here. On MSD Hippocampus, where neither model uses a cascade, Medverse is faster

MSD Hippocampus, posterior (subject hippocampus_001)

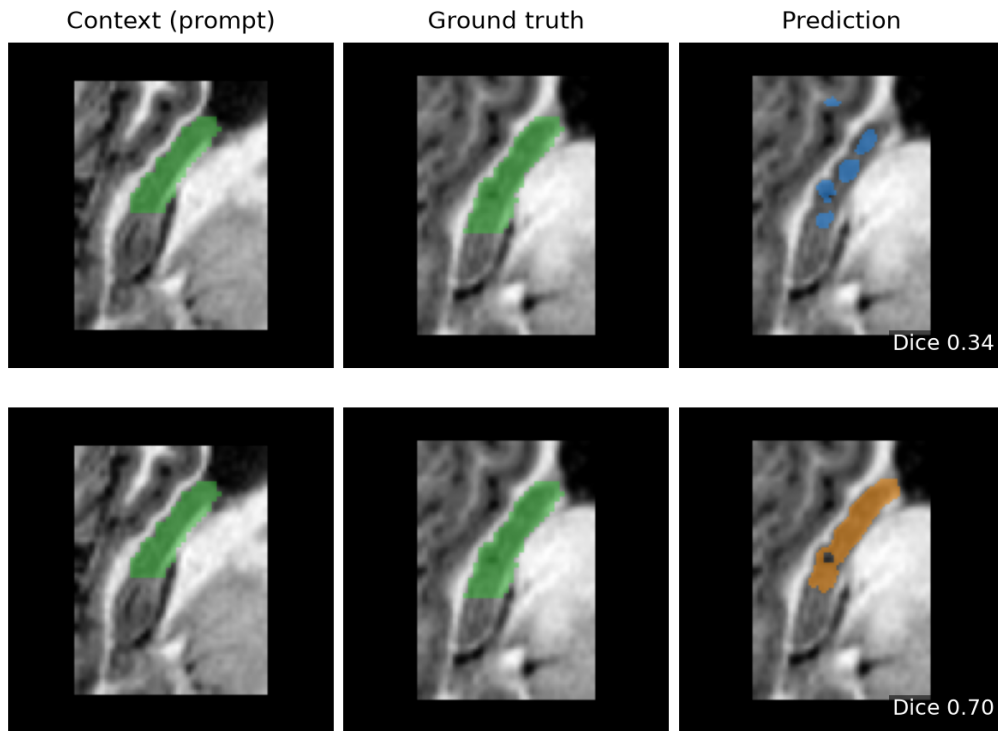


Figure 6: MSD Hippocampus, posterior class, subject hippocampus_001: Ours fragments into disconnected blobs, while Medverse’s native-autoregressive prediction stays a single contiguous region.

(50 vs. 99 ms).

TotalSeg MRI is part of our training data. The large difference there therefore partly reflects training data, not only architecture.

State what Medverse was trained on, including whether MSD Hippocampus is actually among its training classes (currently asserted above without a citation) and whether its training data includes TotalSegmentator. Its matched Dice of 0.055 on TotalSeg MRI is very low; if Medverse’s training data includes TotalSegmentator, check preprocessing (orientation, intensity normalization, spacing) before reporting these numbers.

MSD Prostate TZ (subject prostate_00_adc), common cascade ladder, both in our cascade harness

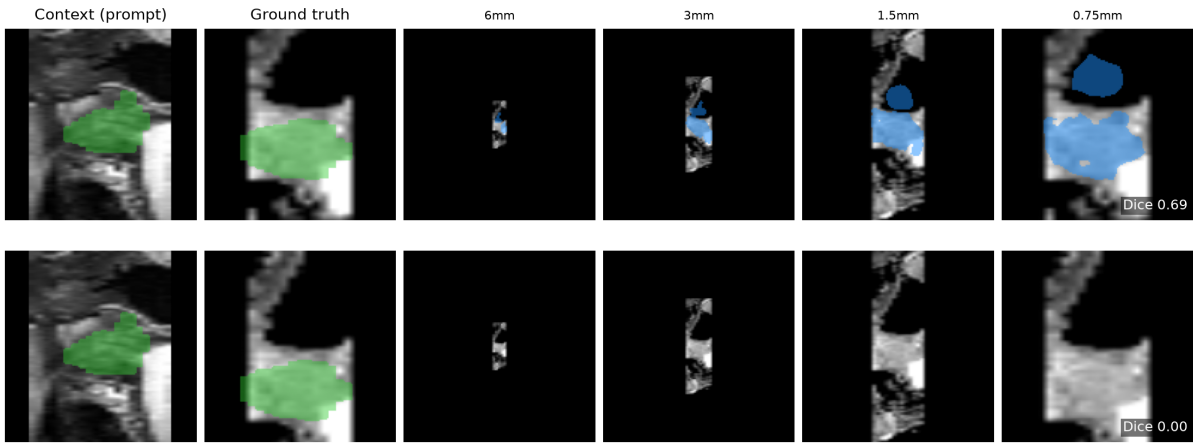


Figure 7: MSD Prostate, TZ class, subject prostate_00_adc, common $[6, 3, 1.5, 0.75]$ mm ladder for both models: Cascade’s prediction grows to fill the zone’s shape while Medverse’s stays empty at every level.

Single-sample illustrations, not aggregate-representative cases (MSD Prostate’s aggregate “prior” Dice is 0.251, not 0.00; the ATLAS v2.0 sample is an extreme outlier the other way, 0.42 vs. an aggregate of 0.020 – both were picked to illustrate a pattern, not summarize the row). All three panels are re-rendered directly from `cascade.run_cascade` on these specific (subject, context, class) triples – not the harness’s own saved diagnostic panels – so the color scheme is consistent (green GT / blue Cascade / orange Medverse) and every level of a shared, common ladder is shown for both models, not just the finest. Forcing Cascade onto a ladder it was not evaluated on for the main tables (MSD Hippocampus’s 3-level ladder; ATLAS v2.0’s $[6, 3, 1.9]$ mm instead of $[3.8, 1.9]$ mm) changes its Dice from the officially reported number (e.g. MSD Hippocampus $0.34 \rightarrow 0.43$, ATLAS v2.0 $0.02 \rightarrow 0.01$) – expected since the crop geometry differs, but worth being explicit that these two figures’ Cascade numbers are not the ones in Table 8. Checkpoints: 2026-09-24_146_cascade_randomfg_predprior_regoff/best.pt (Cascade, released weights for Medverse). The ATLAS v2.0 and common-ladder (subject, context) pairs were not among the sources’ saved figures at all and required re-running `run_cascade` outside the full eval sweep; Medverse’s cascade-mode prediction on the ATLAS v2.0 sample is noticeably unstable across reruns (0.42–0.88 Dice observed across otherwise-identical reruns, plausibly bf16-autocast sensitivity compounding across its two feedback steps) – worth flagging as a finding in its own right, and reason for caution reading any single Medverse-in-cascade number too precisely.

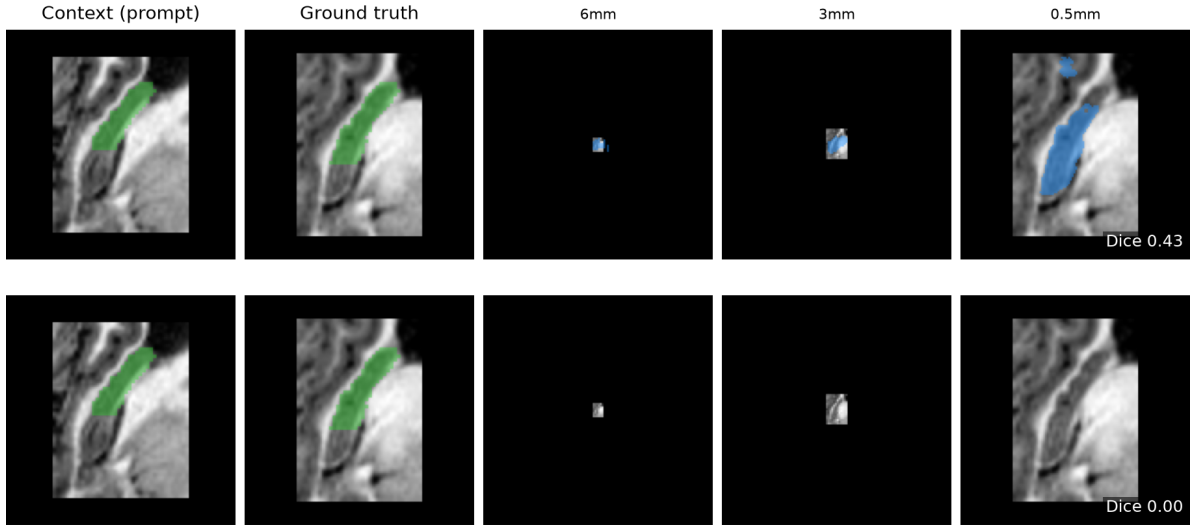


Figure 8: MSD Hippocampus, posterior class, subject hippocampus_001, common $[6, 3, 0.5]$ mm ladder for both models: Medverse’s box is placed correctly at every level but its prediction collapses to empty throughout, consistent with the aggregate drop from 0.691 (no prior) to 0.050 (prior) in Table 15. Cascade’s prediction, forced through the same 3-level ladder for this figure only, is fragmented but present at every level.

Source of the label prior. Table 9 compares two models that differ only in the prior used during training: the model’s own prediction from the previous level, or a perturbed ground-truth mask. At inference, both use their own predicted prior, since no ground truth is available then. On TotalSeg CT, the predicted prior is better on 99 of 117 classes (median difference $+0.043$ Dice) and also improves NSD (0.631 vs. 0.576). FLARE22, also part of the training vocabulary, shows the same direction (0.731 to 0.751). MSD Hippocampus runs a single level, so no prior is passed; its row reflects the two checkpoints, not the prior, and is excluded from the counts below. On the remaining eight datasets outside the training vocabulary (class-level held-out, subsection 4.1; this excludes FLARE22 as well as TotalSeg CT, both already covered above), the predicted prior is better on five and worse on three: Shifts-MS, ATLAS v2.0, and GNC_705. These three, together with ISLES22, are lesion datasets on which both priors score below 0.1; among these four, only ISLES22 favors the predicted prior.

Planned ablations: no prior at all, and no region restriction (full-volume sliding window at fine levels, same checkpoint). Checkpoint 145’s TotalSeg CT/MRI numbers (0.543/0.449 above) are untraceable to any archived run, same gap as checkpoint 146 had before the 2026-09-30 gap-fill (see Table 8’s todo) — 145 was not part of that batch and still needs the equivalent re-run.

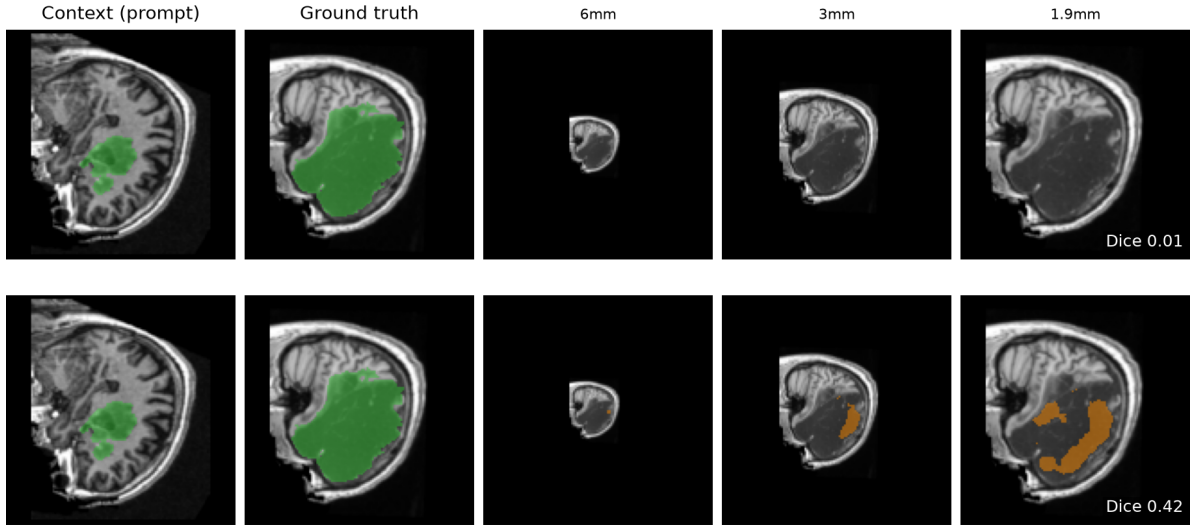


Figure 9: ATLAS v2.0, stroke lesion, subject sub-r004s004, common [6, 3, 1.9] mm ladder for both models: Cascade’s predicted region shrinks to almost nothing by its finest level while Medverse’s prior-conditioned prediction grows to cover much of the lesion. This sample scores far above ATLAS v2.0’s aggregate Prior Dice (0.42 vs. 0.020) and was chosen specifically to show that the prior does not hurt Medverse uniformly, not as a representative case. This particular Dice value is also unstable across reruns of Medverse’s cascade-mode prediction on this exact sample (0.42–0.88 observed, see the todo below) — read it as illustrating the direction of the effect, not as a precise number.

4.4 Synthetic Task Generation

This section evaluates the synthetic task generator, mainly on the four lesion datasets it targets: ISLES22, Shifts-MS, ATLAS v2.0, and GNC_705. They are the most difficult evaluation datasets, since their targets are lesions on MRI with no counterpart among the training classes. Under cascade inference, the comparison widens to all nine held-out datasets used in subsection 4.3.

Synthetic-bank canvas. The baseline is the Synth configuration trained with $p_{\text{synth}} = 0.5$ and the synthetic-bank canvas. It reaches Dice between 0.008 and 0.082 on the four datasets (Table 10). Changing the mix of shape families or p_{synth} with this canvas did not change these results beyond run-to-run variation. The bank’s images are synthesized from label maps of CT scans, whereas all four target datasets are real MRI. This suggested that the canvas, rather than the variety of shapes, limits transfer.

Show the family-mix and p_{synth} runs (appendix table)

Table 9: Macro Dice of models trained with a ground-truth target mask prior or with their own predicted prior.

Dataset	Ground-truth prior	Predicted prior
TotalSeg CT	0.543	0.585
TotalSeg MRI	0.449	0.503
FLARE22	0.731	0.751
MSD Prostate	0.310	0.375
HU_LWK1	0.099	0.152
NasalSeg	0.399	0.547
ISLES22	0.067	0.069
Shifts-MS	0.035	0.031
ATLAS v2.0	0.029	0.026
GNC_705	0.027	0.024

Real-host canvas. To test this, we continued training from the baseline checkpoint in two runs that differ only in the canvas: the synthetic bank (stage 1) or real scans from TotalSegmentator (stage 2). Both use only the scatter-field family, with the same p_{synth} and training budget. The real canvas improves Dice on three of the four datasets: ISLES22 from 0.045 to 0.098, Shifts-MS from 0.011 to 0.058, and GNC_705 from 0.070 to 0.105. ATLAS v2.0 is unchanged (0.033 vs. 0.032). ATLAS v2.0 mostly contains single compact lesions, whereas scatter-field shapes consist of many small components, which may explain why it does not benefit. Absolute Dice remains low on all four datasets.

Confirm that stages 1 and 2 are continued from the baseline checkpoint, and what distinguishes stage 1 from the baseline.

In stage 3, the real-host canvas is extended to all seven shape families and to all TotalSegmentator subjects (CT and MRI, all organs), and training runs longer. This raises Dice on GNC_705 to 0.223, while ISLES22, Shifts-MS, and ATLAS v2.0 all drop slightly below their stage-2 values. Because stage 3 changes three factors at once, the result cannot be attributed to any one of them.

Effect on TotalSegmentator. Table 11 reports TotalSegmentator validation Dice for the same checkpoints, split by modality and class membership. Stages 1 and 2 change it by at most 0.023 in any row (TotalSeg CT, unseen) relative to the baseline, so the gains of stage 2 on the lesion datasets do not reduce accuracy on the training distribution. Stage 3 improves all four values, and the sample-weighted mean rises from 0.434 to 0.505. Given its longer training budget, this gain likely reflects the additional training rather than the synthetic tasks themselves.

Table 10: Dice on the lesion datasets for successive synthetic-task configurations. Stages 1 and 2 differ only in the canvas. Stage 3 adds all shape families, all TotalSegmentator subjects as canvas, and a longer training budget. Bold marks the best configuration per dataset, as in Table 12.

Dataset	Baseline	1) Synthetic bank	2) Real host, scatter	3) Real host, all families
ISLES22	0.050	0.045	0.098	0.087
Shifts-MS	0.008	0.011	0.058	0.051
ATLAS v2.0	0.028	0.033	0.032	0.027
GNC_705	0.082	0.070	0.105	0.223
Mean (sample-weighted) [†]	0.063	0.057	0.084	0.153

[†]Sample-weighted over the four datasets, weighted by each one’s own number of evaluated tasks (46 for Shifts-MS to 1490 for GNC_705, identical across all four configurations). GNC_705 alone is 61% of the total weight, so this mean tracks GNC_705 closely; the unweighted mean (Baseline 0.042, stage 1 0.040, stage 2 0.073, stage 3 0.097) ranks the four configurations the same way, but with a much smaller stage-3 lead.

Cascade inference. Synth + Cascade continues the stage-3 checkpoint with 20 epochs of cascade training. Table 12 compares it with Cascade, which uses no synthetic tasks, and with step-matched Medverse on the nine held-out datasets. Synth + Cascade improves on MSD Hippocampus (0.303 to 0.518) and NasalSeg (0.547 to 0.644), is worse on HU_LWK1 (−0.028) and Shifts-MS (−0.012), and differs by at most 0.011 on the other datasets. On the sample-weighted mean of the five non-lesion datasets, Synth + Cascade is highest (0.615), ahead of Medverse (0.552) and Cascade (0.531); this ordering is driven by MSD Hippocampus and NasalSeg, the two datasets where Cascade trails both other models by the largest margins. It remains below Medverse on MSD Hippocampus, NasalSeg, and Shifts-MS, where Medverse’s step-matched result (0.098) is the best of the three despite this being one of the four lesion datasets that motivated the synthetic-task generator. The two models differ in more than the synthetic tasks: Synth + Cascade also uses register carry and a pooling token, and has a different training history. The differences can therefore not be attributed to the synthetic tasks alone.

The gain on GNC_705 at a single level (0.223, Table 10) does not carry over to cascade inference, where both models score below 0.04. A likely cause is localization: on this dataset, the coarse predictions follow bright regions rather than the lesions, so the region-restricted fine levels process crops that miss the lesion. This illustrates the limitation of region restriction noted in subsection 3.3: small targets that the coarse level mislocalizes cannot be recovered. Figure 10 shows one such case: the lesion is a handful of voxels at the edge of the kidney (yellow circle, added for visibility, on the ground-truth panel), and the region-restricted finest level predicts nothing in that same location (Dice 0.00).

Table 11: TotalSegmentator validation Dice for the checkpoints of Table 10, by modality and class membership, computed the same way as every other results table (macro: per-class mean, then averaged over classes). This differs from an earlier version of this table, whose MRI/CT rows were computed as a flat per-sample mean instead (matching `train.py`’s own internal logging); under that convention TotalSeg MRI scored below TotalSeg CT overall, but splitting seen from unseen first, as here, reverses that (MRI-seen scores above CT-seen at every stage) — a composition effect, not a correction of an error.

	Baseline	1) Synthetic bank	2) Real host, scatter	3) Real host, all families
TotalSeg CT, seen	0.479	0.476	0.474	0.572
TotalSeg CT, unseen	0.326	0.348	0.337	0.385
TotalSeg MRI, seen	0.505	0.500	0.498	0.590
TotalSeg MRI, unseen	0.395	0.379	0.379	0.430
Mean (sample-weighted) [†]	0.434	0.435	0.430	0.505

[†]Sample-weighted over the four rows, weighted by each one’s own number of evaluated tasks (155 for TotalSeg MRI, unseen to 273 for TotalSeg MRI, seen, identical across all four configurations); the unweighted mean (Baseline 0.426, stage 1 0.426, stage 2 0.422, stage 3 0.494) ranks the configurations the same way, though Baseline and stage 1 are within 0.0002 of each other under either weighting and should be read as tied.

Support this further with the containment of the coarse box on GNC_705 across all samples, not just this one case (currently a single qualitative example, re-rendered directly from `cascade.run_cascade` on this one (subject, context, class) triple, not a systematic worst-case search), instead of the reference to `docs/datasets/gnc_kidney_lesions.md`. If the single-level evaluation uses ground-truth-centered crops, the drop may mainly reflect the loss of ground-truth localization; state which is the case.

Final comparison: Synth + Cascade vs. Medverse. Table 13 adds NSD and latency to the Dice comparison above, restricted to our best cascade configuration, Synth + Cascade, against step-matched Medverse. On the sample-weighted mean of the five non-lesion datasets, Synth + Cascade leads on both Dice (0.615 vs. 0.552) and NSD (0.729 vs. 0.631) while being 15× faster (179 vs. 2697 ms per sample), consistent with the 15–17× latency gap already reported for Cascade in subsection 4.3. NSD moves in the same direction as Dice on seven of the nine datasets; the exceptions are Shifts-MS, where Medverse’s Dice lead (0.098 vs. 0.019) is even larger on NSD (0.136 vs. 0.123), and ATLAS v2.0, where Synth + Cascade’s Dice lead (0.032 vs. 0.024) is much larger still on NSD (0.044 vs. 0.012). On the four lesion datasets, absolute scores stay below 0.1 Dice for both models on every one of them, and Medverse is faster only on MSD Hippocampus, the one dataset run at a single level rather than a cascade. ?? in the appendix breaks this comparison down to every individual class on all nine datasets.

Table 12: Dice on the held-out datasets under cascade inference. Medverse: released weights, step-matched autoregressive inference (subsection 4.3). Cascade and Synth + Cascade use the same per-dataset ladder (identical sample counts confirmed for every dataset), so differences are not a ladder confound; Medverse uses its own depth-matched ladder from Table 8.

Dataset	Spacings (mm)	Cascade	Synth + Cascade	Medverse
FLARE22	[6, 3, 2.5]	0.751	0.743	0.359
MSD Prostate	[3, 1.5, 0.75]	0.375	0.372	0.357
MSD Hippocampus	[0.5]	0.303	0.518	0.686
HU_LWK1	[6, 3, 1]	0.152	0.124	0.020
NasalSeg	[1.5, 0.8]	0.547	0.644	0.737
Mean (sample-weighted) [†]		0.531	0.615	0.552
ISLES22	[3, 1.5]	0.069	0.070	0.059
Shifts-MS	[3, 1.5]	0.031	0.019	0.098
ATLAS v2.0	[3.8, 1.9]	0.026	0.032	0.024
GNC_705	[6, 3, 1.2]	0.024	0.035	0.007

[†]Sample-weighted over the five datasets above the second midrule, each weighted by its own number of evaluated tasks (36 for HU_LWK1 to 650 for FLARE22, identical across all three models here); the unweighted mean-of-datasets ranks the three models the same way (Cascade 0.426, Synth + Cascade 0.480, Medverse 0.432). The four lesion datasets below the second midrule are excluded (all three models score below 0.1 on every one of them).

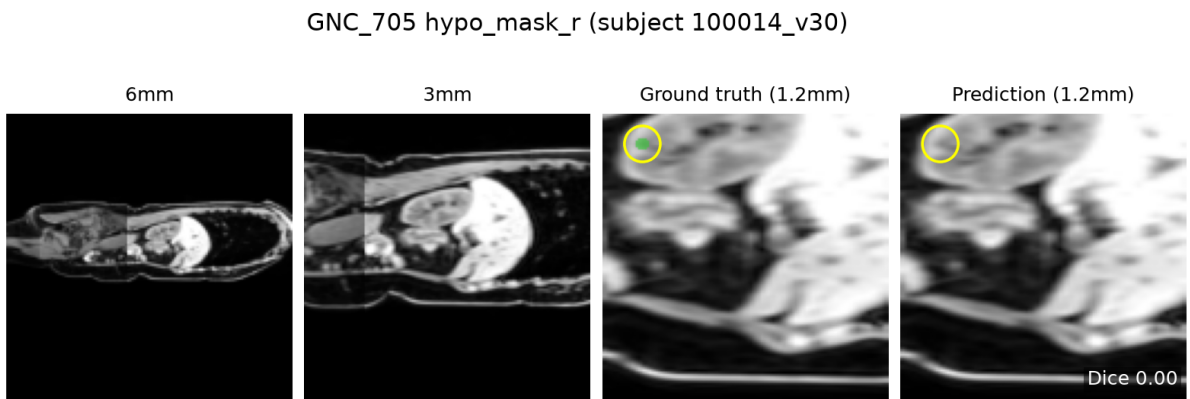


Figure 10: GNC_705, class hypo_mask_r, subject 100014_v30 (Cascade checkpoint), coarse (6 mm) and mid (3 mm) level images shown for context: the lesion (yellow circle) is a handful of voxels at the edge of the kidney; the finest (1.2 mm) level, restricted to its own region, predicts nothing there (Dice 0.00).

Table 13: Final comparison, Synth + Cascade (ours) vs. Medverse: Dice, NSD, and mean latency per sample on all nine held-out datasets. Medverse: released weights, autoregressive inference with its number of steps matched to our number of levels per dataset (subsection 4.3). Bold marks the best value per row for Dice and NSD; latency is bolded only in the sample-weighted mean row.

Dataset	Spacings (mm)	Dice		NSD		Latency (ms)	
		Ours	Medverse	Ours	Medverse	Ours	Medverse
FLARE22	[6, 3, 2.5]	0.743	0.359	0.775	0.333	274	4894
MSD Prostate	[3, 1.5, 0.75]	0.372	0.357	0.314	0.300	278	5013
MSD Hippocampus	[0.5]	0.518	0.686	0.753	0.905	97	50
HU_LWK1	[6, 3, 1]	0.124	0.020	0.049	0.029	407	5336
NasalSeg	[1.5, 0.8]	0.644	0.737	0.791	0.844	106	1886
Mean (sample-weighted) [†]		0.615	0.552	0.729	0.631	179	2697
ISLES22	[3, 1.5]	0.070	0.059	0.086	0.070	145	1894
Shifts-MS	[3, 1.5]	0.019	0.098	0.123	0.136	272	1911
ATLAS v2.0	[3.8, 1.9]	0.032	0.024	0.044	0.012	151	1862
GNC_705	[6, 3, 1.2]	0.035	0.007	0.044	0.005	225	4874

[†]Sample-weighted over the five datasets above the second midrule, each weighted by its own number of evaluated tasks (36 for HU_LWK1 to 650 for FLARE22, identical across both models here); the unweighted mean-of-datasets ranks the two models the same way on every metric (Dice: Synth + Cascade 0.480, Medverse 0.432. NSD: 0.536, 0.482. Latency: 232, 3436 ms). The four lesion datasets below the second midrule are excluded from the means (both models score below 0.1 Dice on every one of them). Dice and NSD are macro (per-class mean, then averaged over classes, as in subsection 4.1); latency is a flat mean over samples.

5 Discussion

This chapter summarizes what the experiments show for each of the three questions raised in section 1, discusses the limitations of the evidence, and outlines open questions.

5.1 Image–Label Fusion

Replacing Medverse’s two U-Nets with a shared encoder and attention layers improved accuracy on classes from the training vocabulary by a large margin, between 0.16 and 0.23 Dice on TotalSegmentator and FLARE22 (Table 6). On new targets, this change alone did not help: early fusion was below Dual U-Net Cross-Attn on MSD Prostate, MSD Hippocampus, and HU_LWK1. Dual attention changed this pattern only partly. Its gains over early fusion were small on in-vocabulary classes and larger on new targets and on the unseen modality, and it was the only variant above Dual U-Net Cross-Attn on the sample-weighted mean over held-out sources. This aggregate advantage does not hold source-by-source: on the five lesion and cavity datasets, Dual U-Net Cross-Attn wins more often (four of five), so the mean is driven by the first five, more anatomical, held-out sources.

One interpretation is that repeated interaction between image and label tokens matters most when the model cannot rely on having seen the class during training and has to extract the target from the context masks. This is consistent with the results, but they do not test it directly. Dual attention also remained well below Dual U-Net Cross-Attn on MSD Prostate and HU_LWK1. Dual U-Net Cross-Attn starts from Medverse’s weights, which were trained on different data; part of its advantage on these datasets may come from that pretraining rather than from its architecture.

Dual attention reached these results with the fewest parameters of the three variants. The GFLOPs of the three variants differ by up to 65%, but their latency is nearly identical, so GFLOPs did not predict latency in this setting.

Check where the time goes (e.g. encoder, data movement) before drawing conclusions about compute.

5.2 Coarse-to-Fine Cascade

When each level’s crop is placed from the previous level’s prediction, Dice still increases from level to level on every dataset (Table 7). On TotalSeg CT, the coarse level localizes

the target well enough for the first, coarsest-to-middle step: a box at its predicted centroid contains 92% of the target’s voxels on average, compared with 97% for a box at the ground-truth centroid, and most of the remaining error at this step arises within the fine crops rather than from localization; the middle-to-finest step is not measured this way.

The clearest advantage over Medverse is speed. Cascade is about 17 times faster than Medverse with the same number of refinement steps, sample-weighted over the six datasets where both latencies are measured. The accuracy comparison depends on how Medverse is run. With matched steps, Cascade is more accurate on seven of the ten datasets with a matched value; against Medverse at its native number of steps, it is more accurate on six of the same ten (FLARE22, HU_LWK1, TotalSeg MRI, ISLES22, ATLAS v2.0, and GNC_705). The efficiency result is thus better supported than the accuracy result. TotalSeg CT is not part of this comparison (subsection 4.3).

Training with the model’s own predictions as prior worked better than training with perturbed ground-truth masks on the training distribution (99 of 117 classes) and was mixed on datasets outside the training vocabulary (five of eight). A plausible explanation is that a model trained on its own predictions sees, during training, the kind of errors it will encounter at inference, which a hand-designed perturbation of the ground truth only approximates. Whether the prior helps at all compared with no prior has not been tested yet. The experiment with Medverse inside our cascade shows that a prior can also hurt: passing Medverse’s coarse prediction forward lowered its Dice on six of seven datasets.

Region restriction makes the cost of the fine levels depend on the size of the target rather than the size of the volume, but it makes the fine levels depend on the coarse localization. GNC_705 shows the consequence: its single-level Dice of 0.223 dropped below 0.04 under cascade inference, likely because the coarse level mislocalizes the small lesions. Small, multi-focal, or low-contrast targets are the cases where this failure is most likely. Possible remedies include larger margins, several boxes per volume for multi-focal targets, or falling back to full-volume processing when the coarse prediction is uncertain.

Revise the GNC_705 paragraph once it is clear whether the single-level evaluation used ground-truth-centered crops (subsection 4.4).

5.3 Synthetic Task Generation

With the synthetic-bank canvas, varying the shapes and the fraction of synthetic tasks had no measurable effect on the lesion datasets. Changing only the canvas from CT bank images to real MRI cases improved three of four lesion datasets without reducing accuracy on TotalSegmentator. This indicates that the image the shapes are painted on mattered more than the variety of shapes, at least when the canvas and the target data differ in modality.

The effect of synthetic tasks under cascade inference was smaller and different. Synth + Cascade improved mainly on MSD Hippocampus and NasalSeg, which are anatomical structures rather than lesions, and not on the lesion datasets that motivated the generator. Since Synth + Cascade also differs from Cascade in other components and in training history, these gains cannot be attributed to the synthetic tasks alone.

Across all configurations, Dice on the lesion datasets stayed low: at most 0.223 at a single level and below 0.08 under cascade inference. In-context segmentation of lesions remains largely unsolved in this setting. Lesions vary in location, number, and shape far more than anatomical structures, so the context examples say less about where the target is. How much of this gap synthetic tasks can close remains open.

5.4 Limitations

Strength of the evidence. All results come from a single training run per configuration, so no variance estimates are available, and differences of a few hundredths of Dice should not be over-interpreted. Several results rest on few classes, such as the two unseen FLARE22 classes, the MSD datasets with two classes each, and HU_LWK1 with one. In the fusion comparison, early fusion and dual attention had not completed their training budget, while Dual U-Net Cross-Attn had converged.

Differences between configurations. The three results sections use different model configurations (Table 3), so their numbers cannot be compared with each other. Within sections, some comparisons change several factors at once: stage 3 of the synthetic-task experiments changes shape families, canvas pool, and training budget together, and Synth + Cascade differs from Cascade in more than the synthetic tasks.

Update if a single final configuration is evaluated before submission.

Missing ablations. The contribution of the label prior and of region restriction has not been isolated: the cascade has not been evaluated without a prior or with full-volume processing at the fine levels. The speedup over Medverse therefore reflects the whole system, not region restriction alone.

Baselines and data. Medverse is the only baseline evaluated empirically. Iris and UniverSeg are compared only by design. The overlap between Medverse’s training data and our evaluation datasets is not fully documented, which affects the comparison on TotalSegmentator, MSD Hippocampus, and NasalSeg. The provenance of the ATLAS v2.0 copy used here has not been confirmed.

Remove the ATLAS sentence once provenance is confirmed or the dataset is dropped.

5.5 Open Questions

The most direct next steps are the missing ablations (no prior, no region restriction), additional seeds, and a single final configuration evaluated on all datasets. Beyond these, three questions remain open. First, the position axis attends over the tokens of all $K + 1$ volumes at once, so its cost grows quadratically with the number of context volumes. How accuracy and cost scale with K has not been studied. Second, region restriction needs a way to handle targets the coarse level cannot localize reliably. Third, synthetic tasks helped on some held-out datasets, but not yet on lesions under cascade inference. Generating tasks that resemble lesions in appearance and distribution, not only in shape, is one direction to explore.

6 Conclusion

This thesis studied in-context segmentation of 3D medical images and addressed three questions: how to combine image and label information, where to spend compute, and how to obtain training data beyond the annotated classes.

For the first question, we proposed dual attention, which keeps image and label tokens separate and lets them interact in every layer. In a comparison trained on TotalSegmentator CT, dual attention improved over merging the two once, mainly on new targets and on an unseen modality, and it did so with the fewest parameters of the compared variants. It was also the only shared-encoder variant above a retrained Medverse architecture on the mean over held-out datasets, although not on every dataset.

For the second question, we proposed a coarse-to-fine cascade in which each level receives the previous level’s prediction as a prior and processes only the region that prediction localizes. The cascade was about 17 times faster than Medverse with the same number of refinement steps. It was more accurate than step-matched Medverse on seven of ten datasets, but the comparison with Medverse at its native depth was mixed. Training with the model’s own predictions as prior worked better than training with perturbed ground-truth masks on the training distribution.

For the third question, we proposed a synthetic task generator that paints procedural shapes onto scans. The modality of the image the shapes are painted on mattered more than the variety of shapes: painting onto real MRI cases improved three of four lesion datasets at a single level without reducing accuracy on TotalSegmentator. Under cascade inference, synthetic tasks helped on held-out anatomical structures but not on lesions.

All results come from single training runs, and several comparisons are not fully controlled; subsection 5.4 lists the main limitations and the ablations still missing. Segmenting lesions from a few examples remains the hardest case in this setting and the clearest direction for future work.

Update the numbers and claims once the pending re-runs and ablations are complete.

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Appendix

Table 15: Per-class Dice and NSD, Synth + Cascade (ours) vs. Medverse, extending the dataset-level results of Table 13 to every class on all nine held-out datasets. Same protocol: Medverse’s number of autoregressive steps matched to our number of levels per dataset (subsection 4.3). Bold marks the better value per row; exact ties are left unbolded (GNC_705, Hyper mask L, NSD: both 0.000).

Dataset	Class	Dice		NSD	
		Ours	Medverse	Ours	Medverse
FLARE22	Adrenal gland L	0.623	0.001	0.806	0.000
	Adrenal gland R	0.260	0.001	0.415	0.000
	Aorta	0.878	0.454	0.929	0.461
	Duodenum	0.500	0.020	0.502	0.029
	Esophagus	0.678	0.004	0.747	0.005
	Gallbladder	0.757	0.019	0.797	0.017
	Inferior vena cava	0.808	0.119	0.813	0.112
	Kidney L	0.910	0.903	0.919	0.875
	Kidney R	0.908	0.915	0.910	0.902
	Liver	0.950	0.945	0.893	0.851
	Pancreas	0.616	0.077	0.624	0.063
	Spleen	0.934	0.755	0.960	0.720
	Stomach	0.832	0.448	0.757	0.294
MSD Prostate	Prostate PZ	0.188	0.075	0.245	0.121
	Prostate TZ	0.557	0.640	0.382	0.480
MSD Hippocampus	Hippocampus anterior	0.562	0.704	0.767	0.902
	Hippocampus posterior	0.473	0.668	0.739	0.907
HU_LWK1	L1 center	0.124	0.020	0.049	0.029
NasalSeg	Maxillary sinus L	0.763	0.861	0.826	0.912
	Maxillary sinus R	0.799	0.822	0.864	0.890
	Nasal cavity L	0.419	0.633	0.685	0.805
	Nasal cavity R	0.443	0.556	0.737	0.740
	Nasopharynx	0.795	0.812	0.843	0.874
ISLES22	Stroke lesion	0.070	0.059	0.086	0.070
Shifts-MS	MS lesion	0.019	0.098	0.123	0.136
ATLAS v2.0	Stroke lesion	0.032	0.024	0.044	0.012

continued on next page

Table 15 – continued from previous page

		Dice		NSD		
Dataset	Class	Ours	Medverse	Ours	Medverse	
GNC_705	Complex L		0.092	0.026	0.106	0.027
	Complex R		0.032	0.010	0.034	0.009
	Complex cyst L		0.038	0.004	0.050	0.003
	Complex cyst R		0.019	0.002	0.032	0.001
	Hyper L		0.053	0.003	0.075	0.001
	Hyper R		0.021	0.003	0.029	0.000
	Hyper cyst L		0.079	0.006	0.102	0.001
	Hyper cyst R		0.008	0.005	0.015	0.000
	Hyper mask L		0.000	0.004	0.000	0.000
	Hypo L		0.035	0.005	0.035	0.003
	Hypo R		0.016	0.009	0.018	0.005
	Hypo mask L		0.033	0.008	0.038	0.004
	Hypo mask R		0.033	0.010	0.036	0.008

Table 14: TotalSegmentator classes used for training (seen) and held out (unseen), with the number of classes per category in parentheses. †: also available on TotalSegmentator MRI. On MRI, 50 classes are available, 31 of them seen.

Category	Seen	Unseen
Organs, abdomen and pelvis (12/5)	adrenal gland L†, colon†, duodenum†, esophagus†, gallbladder†, kidney L†, liver†, pancreas†, small bowel†, spleen†, stomach†, urinary bladder†	adrenal gland R†, kidney R†, prostate†, kidney cyst L, kidney cyst R
Organs, thorax, head, and spine (8/3)	brain†, heart†, spinal cord†, thyroid gland, trachea, lung lower lobe L, lung middle lobe R, lung upper lobe R	lung lower lobe R, lung upper lobe L, left atrial appendage
Muscles (5/5)	autochthon L†, gluteus maximus L†, gluteus medius R†, gluteus minimus L†, iliopsoas R†	autochthon R†, gluteus maximus R†, gluteus medius L†, gluteus minimus R†, iliopsoas L†
Bones, limbs, shoulder, and pelvis (6/5)	clavicle R†, femur L†, hip R†, humerus L†, scapula L†, skull	clavicle L†, femur R†, hip L†, humerus R†, scapula R†
Bones, spine (11/15)	sacrum†; vertebrae C3, C7, T3, T6, T9, T12, L1, L3, L5, S1	vertebrae C1, C2, C4–C6, T1, T2, T4, T5, T7, T8, T10, T11, L2, L4
Bones, ribs and sternum (9/17)	sternum, costal cartilages; ribs L2, L3, L9, L12, R2, R6, R11	ribs L1, L4–L8, L10, L11, R1, R3–R5, R7–R10, R12
Vessels (10/6)	aorta†, inferior vena cava†, portal and splenic vein†, iliac artery L†, iliac vein L†, superior vena cava, pulmonary vein, brachiocephalic trunk, common carotid artery L, subclavian artery L	iliac artery R†, iliac vein R†, common carotid artery R, subclavian artery R, brachiocephalic vein L, brachiocephalic vein R
MRI only (0/4)	–	lung L, lung R, vertebrae (all levels merged), intervertebral discs